

1. For each of the following, give an example of a sequence with that property or prove that no such sequences exist.
  - (a) A Cauchy sequence which has an unbounded subsequence.
  - (b) An unbounded sequence which has a Cauchy sequence.

2. Find the limit of the sequences with general term as given:

(a)  $\frac{2024n^2 + 42}{n^2 - 51}$

(b)  $\frac{\sin(n^{2024})}{\sqrt{n}}$

3. Assume that  $\{a_n\}_{n=1}^{\infty}$  and  $\{b_n\}_{n=1}^{\infty}$  are sequences such that  $a_n \rightarrow A$ ,  $b_n \rightarrow B$ . Suppose that  $b_n \neq 0$  for all  $n$  and  $B \neq 0$ . Show that

$$\frac{a_n}{b_n} \rightarrow \frac{A}{B}.$$

4. Show that the sequence defined by  $a_1 = 2024$  and  $a_n = \sqrt{6 + a_{n-1}}$  for  $n > 1$  is convergent and find its limit.

5. Let  $a_1$  and  $a_2$  be real numbers. Define

$$a_n = \frac{a_{n-1} + a_{n-2}}{2}, \quad n \geq 3.$$

Does  $\{a_n\}_{n=1}^{\infty}$  necessarily converge? Prove it or provide a counterexample.

6. Consider the sequence  $\{a_n\}_{n=1}^{\infty}$  and define a new sequence  $\{b_n\}_{n=1}^{\infty}$  by

$$b_n = \frac{a_1 + \dots + a_n}{n}.$$

- (a) Prove that if  $a_n \rightarrow A$ , then  $b_n \rightarrow A$ .
- (b) Show that the converse does not hold.

7. Let  $\{a_n\}_{n=1}^{\infty}$  be a bounded sequence, and consider the sequence  $\{b_n\}_{n=1}^{\infty}$  defined by

$$b_n = \sup \{a_n, a_{n+1}, a_{n+2}, \dots\}.$$

Show that  $b_n$  converges.

8. Assume that  $0 \leq a \leq b$ . Show that  $\{(a^n + b^n)^{\frac{1}{n}}\}_{n=1}^{\infty}$  converges and find its limit.